



Lecture 15: Course Review

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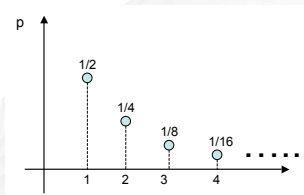
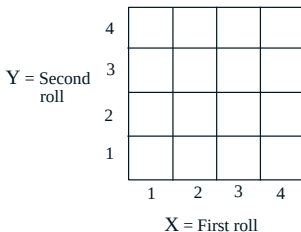
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1. Sample Space and Probability
2. Discrete Random Variables
3. General Random Variables
4. Further Topics on Random Variables
5. Limit Theorems
6. Bernoulli and Poisson Processes

1. Sample Space and Probability
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A probabilistic model

- **Sample space:** The set of all possible outcomes of an experiment.
- **Probability law:** An assignment of a nonnegative number $\mathbf{P}(A)$ to each event A .



Given a samples space Ω :

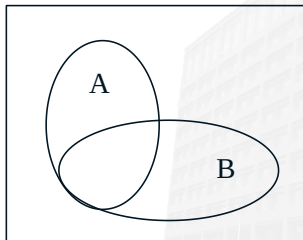
1. **Nonnegativity:** $\mathbf{P}(A) \geq 0$
2. **Normalization:** $\mathbf{P}(\Omega) = 1$
3. **Additivity:** If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$.

If $A_1, A_2, \dots$ are disjoint events, then

$$\mathbf{P}(A_1 \cup A_2 \cup \dots) = \mathbf{P}(A_1) + \mathbf{P}(A_2) + \dots$$

Given events A, B and C :

- If $A \subseteq B$, then $\mathbf{P}(A) \leq \mathbf{P}(B)$
- $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B)$
- $\mathbf{P}(A \cup B) \leq \mathbf{P}(A) + \mathbf{P}(B)$
- $\mathbf{P}(A \cup B \cup C) = \mathbf{P}(A) + \mathbf{P}(A^c \cap B) + \mathbf{P}(A^c \cap B^c \cap C)$



- **Discrete probability law:** If Ω is finite, then each event $A \subseteq \Omega$ can be expressed as

$$A = \{s_1, s_2, \dots, s_n\} \quad s_i \in \Omega.$$

Therefore the probability of the event A is given as

$$\mathbf{P}(A) = \mathbf{P}(s_1) + \mathbf{P}(s_2) + \dots + \mathbf{P}(s_n).$$

- **Discrete uniform probability law:** If all outcomes are equally likely,

$$\mathbf{P}(A) = \frac{|A|}{|\Omega|}.$$

- Given an event B with $\mathbf{P}(B) > 0$, the conditional probability of an event $A \subseteq \Omega$ is given as

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}$$

- $\mathbf{P}(A \mid B)$ is a valid probability law on Ω satisfying
 - Nonnegativity:** $\mathbf{P}(A \mid B) \geq 0$
 - Normalization:** $\mathbf{P}(\Omega \mid B) = 1$
 - Additivity:** $\mathbf{P}(A_1 \cup A_2 \cup \cdots \mid B) = \mathbf{P}(A_1 \mid B) + \mathbf{P}(A_2 \mid B) + \cdots$, where $\{A_i\}$ is a set of disjoint events.
- $\mathbf{P}(A \mid B)$ can also be viewed as a probability law on the restricted universe B .

- Let $A_1, \dots, A_n$ be a set of events such that

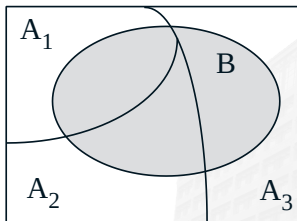
$$\mathbf{P}\left(\bigcap_{i=1}^{n-1} A_i\right) > 0.$$

Then the joint probability of all events is

$$\mathbf{P}\left(\bigcap_{i=1}^n A_i\right) = \mathbf{P}(A_1)\mathbf{P}(A_2 \mid A_1)\mathbf{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbf{P}(A_n \mid \bigcap_{i=1}^{n-1} A_i).$$

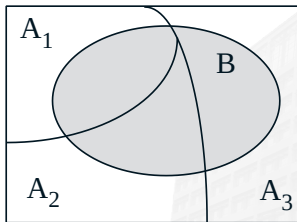
- Let $A_1, \dots, A_n$ be disjoint events that partition Ω . If $\mathbf{P}(A_i) > 0$ for each i , then for any event B ,

$$\mathbf{P}(B) = \sum_{i=1}^n \mathbf{P}(B \cap A_i) = \sum_{i=1}^n \mathbf{P}(B \mid A_i) \mathbf{P}(A_i).$$



- Given a finite partition $A_1, \dots, A_n$ of Ω with $\mathbf{P}(A_i) > 0$, then for each event B with $\mathbf{P}(B) > 0$

$$\mathbf{P}(A_i | B) = \frac{\mathbf{P}(A_i)\mathbf{P}(B | A_i)}{\mathbf{P}(B)} = \frac{\mathbf{P}(A_i)\mathbf{P}(B | A_i)}{\sum_j \mathbf{P}(A_j)\mathbf{P}(B | A_j)}.$$



- Events A and B are **independent** if and only if

$$\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$$

or

$$\mathbf{P}(A \mid B) = \mathbf{P}(A) \quad \text{if } \mathbf{P}(B) > 0.$$

- Events A and B are **conditionally independent** given an event C if and only if

$$\mathbf{P}(A \cap B \mid C) = \mathbf{P}(A \mid C)\mathbf{P}(B \mid C)$$

or

$$\mathbf{P}(A \mid B \cap C) = \mathbf{P}(A \mid C) \quad \text{if } \mathbf{P}(B \cap C) > 0.$$

- Independence $\nLeftrightarrow$ Conditional independence.

- The events $A_1, \dots, A_n$ are **pairwise independent** if for each $i \neq j$

$$\mathbf{P}(A_i \cap A_j) = \mathbf{P}(A_i)\mathbf{P}(A_j).$$

- The events $A_1, \dots, A_n$ are **independent** if

$$\mathbf{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbf{P}(A_i) \quad \forall S \subseteq \{1, 2, \dots, n\}.$$

- Pairwise independence $\nRightarrow$ independence, but independence $\Rightarrow$ pairwise independence.

- **Basic counting principle:** For an m -stage process with n_i choices at stage i ,

$$\# \text{Choices} = n_1 n_2 \cdots n_m.$$

- **Permutations:** k -length sequences drawn from n distinct items without replacement (order is important):

$$\# \text{Sequences} = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

- **Combinations:** Sets with k elements drawn from n distinct items (order within sets is not important):

$$\# \text{Sets} = \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Bonferroni's inequality

(a) Prove that for any two events A and B , we have

$$\mathbf{P}(A \cap B) \geq \mathbf{P}(A) + \mathbf{P}(B) - 1.$$

(b) Generalize to the case of n events $A_1, A_2, \dots, A_n$, by showing that

$$\mathbf{P}(A_1 \cap A_2 \cap \dots \cap A_n) \geq \mathbf{P}(A_1) + \mathbf{P}(A_2) + \dots + \mathbf{P}(A_n) - (n - 1).$$

Solution:

We have $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B)$ and $\mathbf{P}(A \cup B) \leq 1$, which implies part (a).

For part (b), use De Morgan's law to obtain

$$\begin{aligned} 1 - \mathbf{P}(A_1 \cap \cdots \cap A_n) &= \mathbf{P}((A_1 \cap \cdots \cap A_n)^c) \\ &= \mathbf{P}(A_1^c \cup \cdots \cup A_n^c) \\ &\leq \mathbf{P}(A_1^c) + \cdots + \mathbf{P}(A_n^c) \\ &= n - \mathbf{P}(A_1) - \cdots - \mathbf{P}(A_n). \end{aligned}$$

Alice and Bob have $2n + 1$ coins, each coin with probability of heads equal to $1/2$. Bob tosses $n + 1$ coins, while Alice tosses the remaining n coins. Assuming independent coin tosses, show that the probability that after all coins have been tossed, Bob will have gotten more heads than Alice is $1/2$.

Solution:

Let X be the event that after each has tossed n of their coins, Bob has more heads than Alice, let Y be the event that under the same conditions, Alice has more heads than Bob, and let Z be the event that they have the same number of heads. We have

$$\mathbf{P}(X) = \mathbf{P}(Y) \quad \mathbf{P}(Z) = 1 - \mathbf{P}(X) - \mathbf{P}(Y).$$

Furthermore,

$$\mathbf{P}(B|X) = 1, \quad \mathbf{P}(B|Y) = 0, \quad \mathbf{P}(B|Z) = \frac{1}{2}.$$

Use the total probability theorem,

$$\mathbf{P}(B) = 1 \cdot \mathbf{P}(X) + 0 \cdot \mathbf{P}(Y) + \frac{1}{2} \cdot \mathbf{P}(Z) = \frac{1}{2}.$$

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- A random variable is a real-valued function defined on the sample space:

$$X : \Omega \rightarrow \mathbb{R}$$

- The notation $\{X = x\}$ denotes an event:

$$\{X = x\} = \{w \in \Omega \mid X(w) = x\} \subseteq \Omega$$

- The **probability mass function (PMF)** for the random variable X assigns a probability to each event $\{X = x\}$:

$$p_X(x) = \mathbf{P}(\{X = x\}) = \mathbf{P}(\{w \in \Omega \mid X(w) = x\})$$

- Given a random variable X with PMF $p_X(x)$:

$$\mathbf{E}[X] = \sum_x xp_X(x)$$

- Given a derived random variable $Y = g(X)$:

$$\mathbf{E}[g(X)] = \sum_x g(x)p_X(x) = \sum_y yp_Y(y) = \mathbf{E}[Y]$$

$$\mathbf{E}[X^n] = \sum_x x^n p_X(x)$$

- Linearity** of expectation:

$$\mathbf{E}[aX + b] = a\mathbf{E}[X] + b$$

- The expected value of a derived random variable $g(X)$ is

$$\mathbf{E}[g(X)] = \sum_x g(x)p_X(x)$$

The variance of X is calculated as

- $\text{var}(X) = \mathbf{E}[(X - \mathbf{E}[X])^2] = \sum_x (x - \mathbf{E}[X])^2 p_X(x)$
- $\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$
- $\text{var}(aX + b) = a^2 \text{var}(X)$

Note that $\text{var}(X) \geq 0$.

- Let X and Y denote random variables defined on a sample space Ω .
- The joint PMF of X and Y is denoted by

$$p_{X,Y}(x, y) = \mathbf{P}(\{X = x\} \cap \{Y = y\})$$

- The marginal PMFs of X and Y are given respectively as

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

- Let $Z = g(X, Y)$ be a function of two random variables

- **PMF:**

$$p_Z(z) = \sum_{\{(x,y)|g(x,y)=z\}} p_{X,Y}(x, y)$$

- **Expectation:**

$$\mathbf{E}[Z] = \sum_{x,y} g(x, y) p_{X,Y}(x, y)$$

- **Linearity:** Suppose $g(X, Y) = aX + bY + c$

$$\mathbf{E}[g(X, Y)] = a\mathbf{E}[X] + b\mathbf{E}[Y] + c$$

- Conditioning X on an event A with $\mathbf{P}(A) > 0$ results in the PMF:

$$p_{X|A}(x) = \mathbf{P}(\{X = x\} | A) = \frac{\mathbf{P}(\{X = x\} \cap A)}{\mathbf{P}(A)}$$

- Conditioning X on the event $Y = y$ with $\mathbf{P}_Y(y) > 0$ results in the PMF:

$$p_{X|Y}(x|y) = \frac{\mathbf{P}(\{X = x\} \cap \{Y = y\})}{\mathbf{P}(\{Y = y\})} = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

- Multiplication rule:

$$p_{X,Y}(x,y) = p_{X|Y}(x|y)p_Y(y)$$

- Total probability theorem:

$$p_X(x) = \sum_{i=1}^n \mathbf{P}(A_i)p_{X|A_i}(x)$$

$$p_X(x) = \sum_y p_{X|Y}(x|y)p_Y(y)$$

- Given an event A with $\mathbf{P}(A) > 0$

$$\mathbf{E}[X|A] = \sum_x xp_{X|A}(x)$$

- If $p_Y(y) > 0$, then

$$\mathbf{E}[X|\{Y = y\}] = \sum_x xp_{X|Y}(x|y)$$

- Total expectation theorem:** Let $A_1, \dots, A_n$ be a partition of Ω . If $\mathbf{P}(A_i) > 0 \forall i$, then

$$\mathbf{E}[X] = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{E}[X|A_i]$$

- Let X and Y be random variables defined on Ω and let A be an event with $\mathbf{P}(A) > 0$.
- X is independent of A if either of the following holds:

$$p_{X|A}(x) = p_X(x) \quad \forall x$$

$$p_{X,A}(x) = p_X(x)\mathbf{P}(A) \quad \forall x$$

- X and Y are independent if either of the following holds:

$$p_{X|Y}(x|y) = p_X(x) \quad \forall x \quad \forall y$$

$$p_{X,Y}(x, y) = p_X(x)p_Y(y) \quad \forall x \quad \forall y$$

If X and Y are independent, then the following holds:

- If g and h are real-valued functions, then $g(X)$ and $h(Y)$ are independent.
- $\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$ (inverse is not true)
- $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$

Given independent random variables $X_1, \dots, X_n$,

$$\text{var}(X_1 + X_2 + \dots + X_n) = \text{var}(X_1) + \text{var}(X_2) + \dots + \text{var}(X_n)$$

Due to a software glitch, the addresses correspond to the names are in a totally random fashion. You write a holiday card to each of your friends and send it to the (software-corrupted) address. What is the probability that at least one of your friends will get the correct card?



Solution:

Let A_k be the event that the k th card is sent to the correct address.

We have for any k, j, i

$$\mathbf{P}(A_k) = \frac{1}{n} = \frac{(n-1)!}{n!},$$

$$\mathbf{P}(A_k \cap A_j) = \mathbf{P}(A_k)\mathbf{P}(A_j|A_k) = \frac{1}{n} \cdot \frac{1}{n-1} = \frac{(n-2)!}{n!},$$

$$\mathbf{P}(A_k \cap A_j \cap A_i) = \frac{(n-3)!}{n!},$$

etc., and

$$\mathbf{P}(\cap_{k=1}^n A_k) = \frac{1}{n!}.$$

Solution:

Applying the inclusion-exclusion formula,

$$\begin{aligned}\mathbf{P}(\cup_{k=1}^n A_k) &= \binom{n}{1} \frac{(n-1)!}{n!} - \binom{n}{2} \frac{(n-2)!}{n!} + \cdots + (-1)^{n-1} \frac{1}{n!} \\ &= 1 - \frac{1}{2!} + \cdots + (-1)^{n-1} \frac{1}{n!}.\end{aligned}$$

A coin that has probability of heads equal to p is tossed successively and independently until a head comes twice in a row or a tail comes twice in a row. Find the expected value of the number of tosses X .

Solution:

Let H_k (or T_k) be the event that a head (or a tail) comes at the k th toss, and let p (respectively, q) be the probability of H_k (respectively, T_k). Since H_1 and T_1 form a partition of the sample space, and $\mathbf{P}(H_1) = p$ and $\mathbf{P}(T_1) = q$, we have

$$\mathbf{E}[X] = p\mathbf{E}[X|H_1] + q\mathbf{E}[X|T_1].$$

Use again the total expectation theorem,

$$\mathbf{E}[X|H_1] = p\mathbf{E}[X|H_1 \cap H_2] + q\mathbf{E}[X|H_1 \cap T_2] = 2p + q(1 + \mathbf{E}[X|T_1]).$$

where we have used the fact

$$\mathbf{E}[X|H_1 \cap H_2] = 2.$$

Solution:

Similarly,

$$\mathbf{E}[X|T_1] = 2q + p(1 + \mathbf{E}[X|H_1]).$$

Combining the above two relations, and using $p + q = 1$, we obtain

$$\begin{aligned}\mathbf{E}[X|T_1] &= \frac{2 + p^2}{1 - pq}, \\ \mathbf{E}[X|H_1] &= \frac{2 + q^2}{1 - pq}.\end{aligned}$$

Thus

$$\mathbf{E}[X] = p \cdot \frac{2 + q^2}{1 - pq} + q \cdot \frac{2 + p^2}{1 - pq} = \frac{2 + pq}{1 - pq}.$$

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For a continuous r.v. with probability density function $f_X(x)$,

$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$

$$\mathbf{P}(x \in A) = \int_A f_X(x) dx$$

Properties:

- Nonnegativity:

$$f_X(x) \geq 0 \quad \forall x$$

- Normalization:

$$\int_{-\infty}^{\infty} f_X(x) dx = 1$$

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

$$\mathbf{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

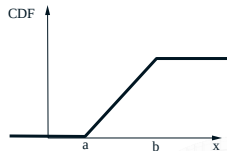
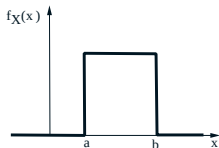
$$\mathbf{E}[aX + b] = a\mathbf{E}[X] + b$$

$$\begin{aligned} \text{var}(X) &= \int_{-\infty}^{\infty} (x - \mathbf{E}[X])^2 f_X(x) dx \\ &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2 \end{aligned}$$

$$\text{var}(aX + b) = a^2 \text{var}(X)$$

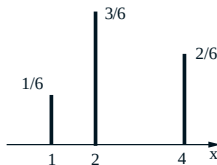
Cumulative distribution function (CDF)

- $F_X(x) = \mathbf{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$



- Also for discrete random variables:

$$F_X(x) = \mathbf{P}(X \leq x) = \sum_{k \leq x} p_X(k)$$



If X is a uniform random variable over the interval $[a, b]$:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & \text{if } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

$$F_X(x) = \begin{cases} 0 & \text{if } a \leq x \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{otherwise} \end{cases}$$

$$\mathbf{E}[X] = \frac{b+a}{2}$$

$$\text{var}(X) = \frac{(b-a)^2}{12}$$

X is an exponential random variable with parameter λ :

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

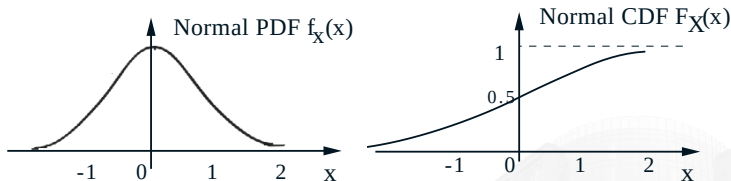
$$F_X(x) = \begin{cases} 1 - e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\mathbf{E}[X] = \frac{1}{\lambda}$$

$$\text{var}(X) = \frac{1}{\lambda^2}$$

Memoryless property: Given that $X > t$, $X - t$ is an exponential r.v. with parameter λ .

- Standard normal $N(0, 1)$: $f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$



- General normal $N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
- If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, $Y \sim N(a\mu + b, a^2\sigma^2)$.

- Joint PDF of two continuous random variables X and Y : $f_{X,Y}(x, y)$

$$\mathbf{P}(A) = \iint_A f_{X,Y}(x, y) dx dy$$

- Marginal PDF

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$$

- Expectation

$$\mathbf{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy$$

- Joint CDF

$$F_{X,Y}(x, y) = \mathbf{P}(X \leq x, Y \leq y)$$

- Two continuous random variables X and Y are independent if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y), \quad \text{for all } x, y.$$

- If X and Y are independent, then

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

- Furthermore, for any function g and h ,

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)]$$

- If X and Y are independent, then

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

- The conditional expectation of X given that A is defined by

$$\mathbf{E}[X|A] = \int_{-\infty}^{\infty} x f_{X|A}(x) dx.$$

- For a function $g(X)$, we have

$$\mathbf{E}[g(X)|A] = \int_{-\infty}^{\infty} g(x) f_{X|A}(x) dx.$$

- **Total expectation theorem:** Let $A_1, A_2, \dots, A_n$ be disjoint events that form a partition of the sample space, and $\mathbf{P}(A_i) > 0$ for all i . Then,

$$\mathbf{E}[X] = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{E}[X|A_i].$$

- Given continuous random variables X and Y

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_Y(y) f_{X|Y}(x|y) dy$$

- Continuous expectation

$$\mathbf{E}[X|Y = y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx$$

$$\mathbf{E}[g(X)|Y = y] = \int_{-\infty}^{\infty} g(x) f_{X|Y}(x|y) dx$$

$$\mathbf{E}[g(X, Y)|Y = y] = \int_{-\infty}^{\infty} g(x, y) f_{X|Y}(x|y) dx$$

- Total expectation theorem

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} \mathbf{E}[X|Y = y] f_Y(y) dy$$

$$\mathbf{E}[g(X)] = \int_{-\infty}^{\infty} \mathbf{E}[g(X)|Y = y] f_Y(y) dy$$

$$\mathbf{E}[g(X, Y)] = \int_{-\infty}^{\infty} \mathbf{E}[g(X, Y)|Y = y] f_Y(y) dy$$

- X, Y continuous r.v., N discrete r.v., A an event

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x)f_X(x)}{f_Y(y)} = \frac{f_{Y|X}(y|x)f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y|t)f_X(t) dt}$$

$$\mathbf{P}(A | Y = y) = \frac{\mathbf{P}(A)f_{Y|A}(y)}{f_Y(y)} = \frac{\mathbf{P}(A)f_{Y|A}(y)}{f_{Y|A}(y)\mathbf{P}(A) + f_{Y|A^c}(y)\mathbf{P}(A^c)}$$

$$\mathbf{P}(N = n | Y = y) = \frac{p_N(n)f_{Y|N}(y|n)}{f_Y(y)} = \frac{p_N(n)f_{Y|N}(y|n)}{\sum_i p_N(i)f_{Y|N}(y|i)}$$

The coordinate X and Y of a point are independent zero mean normal random variables with common variance σ^2 . Given that the point is at a distance of at least c from the origin, find the conditional joint PDF of X and Y .



Solution:

Let C denote the event that $X^2 + Y^2 \geq c^2$. The probability $\mathbf{P}(C)$ can be calculated using the polar coordinate, as follows:

$$\begin{aligned}\mathbf{P}(C) &= \frac{1}{2\pi\sigma^2} \int_0^{2\pi} \int_c^\infty r \exp(-r^2/2\sigma^2) dr d\theta \\ &= \frac{1}{\sigma^2} \int_c^\infty r \exp(-r^2/2\sigma^2) dr \\ &= \exp(-c^2/2\sigma^2).\end{aligned}$$

Therefore, for $(x, y) \in C$,

$$f_{X,Y|C}(x, y) = \frac{f_{X,Y}(x, y)}{\mathbf{P}(C)} = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{1}{2\sigma^2}(x^2 + y^2 - c^2)\right).$$

Let X and Y be independent continuous random variables with PDFs f_X and f_Y , respectively, and let $Z = X + Y$.

- (a) Show that $f_{Z|X}(z|x) = f_Y(z - x)$
- (b) Assume that X and Y are exponential distributed with parameter λ . Find the conditional PDF of X , given that $Z = z$.

Solution:

(a) We have

$$\begin{aligned}\mathbf{P}(Z \leq z|X = x) &= \mathbf{P}(X + Y \leq z|X = x) \\ &= \mathbf{P}(x + Y \leq z|X = x) \\ &= \mathbf{P}(x + Y \leq z) \\ &= \mathbf{P}(Y \leq z - x).\end{aligned}$$

By differentiating both sides with respect to z , the result follows.

Solution:

(b) We have, for $0 \leq x \leq z$,

$$\begin{aligned} f_{X|Z}(x|z) &= \frac{f_{Z|X}(z|x)f_X(x)}{f_Z(z)} \\ &= \frac{f_Y(z-x)f_X(x)}{f_Z(z)} \\ &= \frac{\lambda \exp(-\lambda(z-x))\lambda \exp(-\lambda x)}{f_Z(z)} \\ &= \frac{\lambda^2 \exp(-\lambda z)}{f_Z(z)}. \end{aligned}$$

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2. Discrete Random Variables
3. General Random Variables
- 4. Further Topics on Random Variables**
5. Limit Theorems
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- PDF of a function of a r.v. with known PDF: $Y = g(X)$
- Two-step procedure:
 - Get CDF of Y

$$F_Y(y) = \mathbf{P}(Y \leq y) = \mathbf{P}(g(X) \leq y) = \int_{x|g(x) \leq y} f_X(x) dx$$

- Differentiate to get $f_Y(y) = \frac{dF_Y}{dy}(y)$
- Special case: if $Y = g(X) = aX + b$

$$f_Y(b) = \frac{1}{|a|} f_X\left(\frac{x-b}{a}\right)$$

- If $W = X + Y$ with X and Y independent
- Discrete case:

$$p_W(w) = \sum_x p_X(x)p_Y(w - x)$$

- Continuous case:

$$f_W(w) = \int_{-\infty}^{\infty} f_X(x)f_Y(w - x) dx$$

- **Definition:** $\text{cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])]$.
 - X and Y are uncorrelated if $\text{cov}(X, Y) = 0$.
- **Zero-mean case:** $\text{cov}(X, Y) = \mathbf{E}[XY]$
- $\text{cov}(X, Y) = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y]$
- $\text{cov}(X, X) = \text{var}(X)$
- $\text{cov}(X, aY + b) = a \cdot \text{cov}(X, Y)$
- $\text{cov}(X, Y + Z) = \text{cov}(X, Y) + \text{cov}(X, Z)$
- independent $\implies \text{cov}(X, Y) = 0$ (converse is not true)

- Dimensionless version of covariance:

$$\begin{aligned}\rho &= \mathbf{E} \left[\frac{(X - \mathbf{E}[X])}{\sigma_X} \cdot \frac{(Y - \mathbf{E}[Y])}{\sigma_Y} \right] \\ &= \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}\end{aligned}$$

- $-1 \leq \rho \leq 1$
- $|\rho| = 1 \Leftrightarrow (X - \mathbf{E}[X]) = c(Y - \mathbf{E}[Y])$ (linearly related)
- Independent $\Rightarrow \rho = 0$ (converse is not true)

- $\mathbf{E}[X|Y = y] = f(y)$ is a number
- $\mathbf{E}[X|Y] = f(Y)$ is a random variable
- **Law of iterated expectations:**

$$\begin{aligned}\mathbf{E}[\mathbf{E}[X|Y]] &= \sum_y \mathbf{E}[X|Y = y]p_Y(y) \\ &= \mathbf{E}[X]\end{aligned}$$

- We introduce the random variable

$$\text{var}(X|Y) = \mathbf{E}[(X - \mathbf{E}[X | Y])^2 | Y] = \mathbf{E}[\tilde{X}^2 | Y],$$

- Using the fact $\mathbf{E}[\tilde{X}] = 0$

$$\text{var}(\tilde{X}) = \mathbf{E}[\tilde{X}^2] = \mathbf{E}[\mathbf{E}[\tilde{X}^2 | Y]] = \mathbf{E}[\text{var}(X | Y)],$$

- **Law of total variance:**

$$\text{var}(X) = \text{var}(\mathbf{E}[X|Y]) + \mathbf{E}[\text{var}(X|Y)]$$

- The transform associated with a random variable X

$$M_X(s) = \mathbf{E}[e^{sX}]$$

- not necessarily defined for all (complex) s
- always well defined for $\text{Re}(s) = 0$

- For discrete X :

$$M_X(s) = \sum_x p_X(x) e^{sx}.$$

- For continuous X :

$$M_X(s) = \int_{-\infty}^{\infty} f_X(x) e^{sx} dx.$$

- Consider the sum

$$Y = X_1 + \cdots + X_N ,$$

where N is a random variable that takes integer values, and $X_1, X_2, \dots$ are identically distributed random variables.

- Assume that $N, X_1, X_2, \dots$ are independent.
- Then

$$\mathbf{E}[Y] = \mathbf{E}[N] \mathbf{E}[X]$$

$$\text{var}(Y) = \mathbf{E}[N] \text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N)$$

$$M_Y(s) = M_N(\log M_X(s))$$

Cauchy random variable

- (a) Let X be a random variable that is uniformly distributed between $-1/2$ and $1/2$. Show that the PDF of $Y = \tan(\pi X)$ is

$$f_Y(y) = \frac{1}{\pi(1 + y^2)}$$

- (b) Let Y be a Cauchy random variable. Find the PDF of the random variable X , which is equal to the angle between $-\pi/2$ and $\pi/2$ whose tangent is Y .

Solution:

(a) Note that Y is a continuous, strictly monotonically increasing function of X , which takes values between $-\infty$ and ∞ , as X ranges over $[-1/2, 1/2]$. We have

$$\begin{aligned} F_Y(y) &= \mathbf{P}(Y \leq y) \\ &= \mathbf{P}(\tan(\pi X) \leq y) \\ &= \mathbf{P}(\pi X \leq \tan^{-1} y) \\ &= \frac{1}{\pi} \tan^{-1} y + \frac{1}{2}. \end{aligned}$$

By differentiating, we have for all y ,

$$f_Y(y) = \frac{1}{\pi(1+y^2)}.$$

Solution:

(b) We have for $-\pi/2 \leq x \leq \pi/2$,

$$\begin{aligned}\mathbf{P}(X \leq x) &= \mathbf{P}(\tan^{-1} Y \leq x) \\ &= \mathbf{P}(Y \leq \tan x) \\ &= \int_{-\infty}^{\tan x} \frac{1}{1+y^2} dy \\ &= \frac{1}{\pi} \left(x + \frac{\pi}{2} \right).\end{aligned}$$

For $x < -\pi/2$, we have $\mathbf{P}(X \leq x) = 0$, and for $\pi/2 < x$, we have $\mathbf{P}(X \leq x) = 1$. Taking the derivative of the CDF $\mathbf{P}(X \leq x)$, we find that X is uniformly distributed in $[-\pi/2, \pi/2]$.

Let X and Y be independent random variables that are uniformly distributed on the interval $[0, 1]$. What is the PDF of the random variable $Z = Y/X$?

Solution:

We consider separately the case $0 \leq z \leq 1$ and $z > 1$, we have

$$F_Z(z) = \mathbf{P}\left(\frac{Y}{X} \leq z\right) = \begin{cases} z/2, & \text{if } 0 \leq z \leq 1, \\ 1 - 1/(2z), & \text{if } z > 1, \\ 0, & \text{otherwise.} \end{cases}$$

By differentiating, we obtain

$$f_Z(z) = \begin{cases} 1/2, & \text{if } 0 \leq z \leq 1, \\ 1/(2z^2), & \text{if } z > 1, \\ 0, & \text{otherwise.} \end{cases}$$

We toss n times a biased coin whose probability of heads, denoted by q , is the value of a random variable Q with given mean μ and positive variance σ^2 . Let X_i be a Bernoulli variable that models the outcomes of the i th toss. We assume that $X_1, \dots, X_n$ are conditionally independent, given $Q = q$. Let X be the number of heads obtained in the n tosses.

- (a) Use the law of iterated expectation to find $\mathbf{E}[X_i]$ and $\mathbf{E}[X]$.
- (b) Find $\text{cov}(X_i, X_j)$. Are $X_1, \dots, X_n$ independent?

Solution:

(a) We have, from the law of iterated expectation and the fact $\mathbf{E}[X_i|Q] = Q$,

$$\mathbf{E}[X_i] = \mathbf{E}[\mathbf{E}[X|Q_i]] = \mathbf{E}[Q] = \mu.$$

Since $X = X_1 + \cdots + X_n$, it follows that

$$\mathbf{E}[X] = \mathbf{E}[X_1] + \cdots + \mathbf{E}[X_n] = n\mu.$$

Solution:

(b) We have, for $i \neq j$,

$$\mathbf{E}[X_i X_j | Q] = \mathbf{E}[X_i | Q] \mathbf{E}[X_j | Q] = Q^2,$$

and

$$\mathbf{E}[X_i X_j] = \mathbf{E}[\mathbf{E}[X_i X_j | Q]] = \mathbf{E}[Q^2].$$

Thus

$$\text{cov}(X_i X_j) = \mathbf{E}[X_i X_j] - \mathbf{E}[X_i] \mathbf{E}[X_j] = \mathbf{E}[Q^2] - \mu^2 = \sigma^2.$$

Since $\text{cov}(X_i, X_j) > 0$, $X_1, \dots, X_n$ are not independent.

Use transforms to show that the sum of a Poisson-distributed number of independent, identically distributed Bernoulli random variables is Poisson.



Solution:

Let N be a Poisson-distributed random variable with parameter λ . Let X_i , $i = 1, \dots, N$, be independent Bernoulli random variables with parameter p , and let

$$L = X_1 + \dots + X_N.$$

The transform associated with L is obtained by the transform associated with N

$$M_N(s) = e^{\lambda(e^s - 1)},$$

and replacing each occurrence of e^s by the transform associated with X , which is

$$M_X = 1 - p + pe^s.$$

We obtain

$$M_L(s) = e^{\lambda(1-p+pe^s-1)} = e^{\lambda p(e^s-1)}.$$

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- **Markov inequality:** If a random variable X can only take non-negative values, then

$$\mathbf{P}(X \geq a) \leq \frac{\mathbf{E}[X]}{a}, \quad \text{for all } a > 0$$

- **Chebyshev's inequality:** If a random variable X has a finite mean μ and variance σ^2 , then

$$\mathbf{P}(|X - \mu| \geq c) \leq \frac{\sigma^2}{c^2}$$

- **Chernoff's inequality:** If a random variable X has a transform function $M_X(s)$, then

$$\mathbf{P}(X \geq a) \leq e^{-\max_{s \geq 0} (sa - \ln M_X(s))}$$

- Convergence of the sample mean
- $X_1, X_2, \dots$ i.i.d. finite mean μ and variance σ^2

$$M_n = \frac{X_1 + \dots + X_n}{n}$$

$$\mathbf{P}(|M_n - \mu| \geq \epsilon) \leq \frac{\text{var}(M_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

- M_n converges in probability to μ

- $X_1, \dots, X_n$ i.i.d., finite variance σ^2
- “Standardized” $S_n = X_1 + \dots + X_n$:

$$Z_n = \frac{S_n - \mathbf{E}[S_n]}{\sigma_{S_n}} = \frac{S_n - n\mathbf{E}[X]}{\sqrt{n}\sigma}$$

$$- \mathbf{E}[Z_n] = 0, \quad \text{var}(Z_n) = 1$$

- Let Z be a standard normal random variable
- **Theorem:** For every c :

$$\mathbf{P}(Z_n \leq c) \rightarrow \mathbf{P}(Z \leq c)$$

- $\mathbf{P}(Z \leq c)$ is the standard normal CDF, $\Phi(c)$, available from the normal tables.

The strong law of large numbers

- Let $X_1, X_2, \dots$ be a sequence of independent identically distributed random variables with mean μ . Then

$$M_n = (X_1 + X_2 + \dots + X_n)/n$$

converges to μ , **with probability 1**, in the sense that

$$\mathbf{P}\left(\lim_{n \rightarrow \infty} \frac{X_1 + \dots + X_n}{n} = \mu\right) = 1.$$

We load on a plane 100 packages whose weights are independent random variables that are uniformly distributed between 5 and 50 pounds. What is the probability that the total weight will exceed 3000 pounds?



Solution:

The mean and the variance of the weight of a single package are

$$\mu = \frac{5 + 50}{2} = 27.5, \quad \sigma^2 = \frac{(50 - 5)^2}{12} = 168.75.$$

We calculate the normalized value

$$z = \frac{3000 - 100 \cdot 27.5}{\sqrt{168.75 \cdot 100}} = \frac{250}{129.9} = 1.92,$$

and use the standard normal tables to obtain the approximation

$$\mathbf{P}(S_{100} \leq 3000) \approx \Phi(1.92) = 0.9726.$$

The desired probability is

$$\mathbf{P}(S_{100} > 3000) \approx 1 - 0.9726 = 0.0274.$$

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- Define the Bernoulli process as a sequence $X_1, X_2, \dots$ of independent Bernoulli random variables X_i with

$$\mathbf{P}(\text{success}) = \mathbf{P}(X_i = 1) = p$$

$$\mathbf{P}(\text{failure}) = \mathbf{P}(X_i = 0) = 1 - p$$

for each i .

- Independence:** For any given time n , the sequence of $X_{n+1}, X_{n+2}, \dots$ is also a Bernoulli process, and is independent from $X_1, \dots, X_n$.
- Memorylessness:** Let n be a given time and let $\bar{T}$ be the time of the first success after time n . Then $\bar{T} - n$ has a geometric distribution with parameter p , and is independent of the random variables $X_1, \dots, X_n$.

- The time Y_k of the k th success is equal to the sum $Y_k = T_1 + T_2 + \cdots + T_k$ of k independent identically distributed geometric random variables.
- The mean and variance of Y_k

$$\mathbf{E}[Y_k] = \mathbf{E}[T_1] + \cdots + \mathbf{E}[T_k] = \frac{k}{p},$$

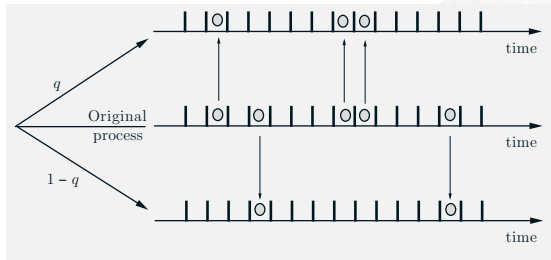
$$\text{var}(Y_k) = \text{var}(T_1) + \cdots + \text{var}(T_k) = \frac{k(1-p)}{p^2}.$$

- The PMF of Y_k is given by

$$p_{Y_k}(t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, \quad t = k, k+1, \dots$$

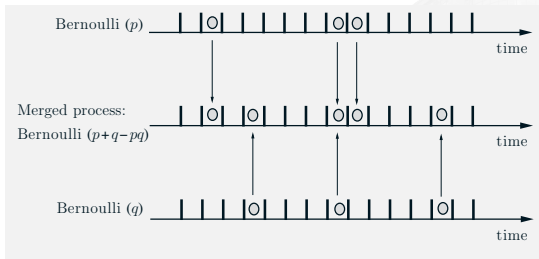
Splitting of a Bernoulli process

- Whenever there is an arrival, we choose to either keep it (with probability q), or to discard it (with probability $1 - q$).
- Both the process of arrivals that are kept and the process of a discarded arrival are Bernoulli processes.



Merging of a Bernoulli process

- In a reverse situation, we start with two independent Bernoulli processes (with parameters p and q respectively). An arrival is recorded in the merged process if and only if there is an arrival in at least one of the two original processes.
- The merged process is Bernoulli, with success probability $p+q-pq$ at each time step.



- **Time homogeneity**

$$\mathbf{P}(k, \tau) = \mathbf{P}(k \text{ arrivals in interval of duration } \tau)$$

- **Independence**

Numbers of arrivals in disjoint time intervals are **independent**.

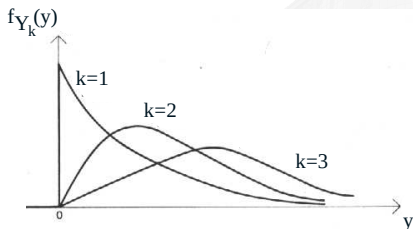
- **Small interval probabilities**

$$P(k, \delta) \approx \begin{cases} 1 - \lambda\tau + o(\tau), & \text{if } k = 0; \\ \lambda\tau + o_1(\tau), & \text{if } k = 1; \\ o_k(\tau), & \text{if } k > 1. \end{cases}$$

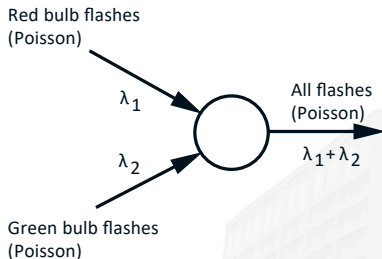
The k th arrival time

- The time Y_k of the k th success is equal to the sum $Y_k = T_1 + T_2 + \cdots + T_k$ of k independent identically distributed exponential random variables.
- Time of the k th arrival, Y_k : **Erlang** distribution

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \quad y \geq 0$$



- Sum of independent Poisson **random variables** is Poisson
- Merging of independent Poisson **processes** is Poisson



- What is the probability that the next arrival comes from the first process?

- Let $N, X_1, X_2, \dots$ be independent random variables, where N takes nonnegative integer values.
- Let $Y = X_1 + \dots + X_N$ for positive values of N , and let $Y = 0$ when $N = 0$.

X_i	N	Y
Bernoulli, p	binomial, (m, q)	binomial, (m, pq)
Bernoulli, p	Poisson, λ	Poisson, λp
geometric, p	geometric, q	geometric, pq
exponential, λ	geometric, q	exponential, λq

Let X_1 and X_2 be independent and exponentially distributed, with parameters λ_1 and λ_2 , respectively. Find the expected value of $\max\{X_1, X_2\}$.

Solution:

Consider two independent Poisson processes with rates λ_1 and λ_2 , respectively. We interpret X_1 as the first arrival time in the first process, and X_2 as the first arrival time in the second process. Let $T = \min\{X_1, X_2\}$ be the first time when one of the processes registers an arrival. Let $S = \max\{X_1, X_2\} - T$ be the additional time until both have registered an arrival. Since the merged process is Poisson with rate $\lambda_1 + \lambda_2$, we have

$$\mathbf{E}[T] = \frac{1}{\lambda_1 + \lambda_2}.$$

Solution:

Concerning S , there are two cases to consider.

(i) The first arrival comes from the first process; this happens with probability $\lambda_1/(\lambda_1 + \lambda_2)$. We then have to wait for an arrival from the second process, which takes $1/\lambda_2$ time on the average.

(ii) The first arrival comes from the second process; this happens with probability $\lambda_2/(\lambda_1 + \lambda_2)$. We then have to wait for an arrival from the first process, which takes $1/\lambda_1$ time on the average. Thus

$$\begin{aligned}\mathbf{E}[\max\{X_1, X_2\}] &= \frac{1}{\lambda_1 + \lambda_2} + \frac{\lambda_1}{\lambda_1 + \lambda_2} \cdot \frac{1}{\lambda_2} + \frac{\lambda_2}{\lambda_1 + \lambda_2} \cdot \frac{1}{\lambda_1} \\ &= \frac{1}{\lambda_1 + \lambda_2} \left(1 + \frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1} \right).\end{aligned}$$

Arrival of two types of jobs, A and B, are independent Poisson processes with parameters $\lambda_A = 3$ and $\lambda_B = 4$ per minute.

- (a) What is the mean, variance, and PMF of the total number of jobs that arrive within a given three-minute interval?
- (b) We are told that during a 10-minute interval, exactly 10 new jobs arrived. What is the probability that exactly 3 of them are of type A?

Solution:

(a) The total arrival process corresponds to the merging of two independent Poisson processes, and is therefore Poisson with rate $\lambda = \lambda_A + \lambda_B = 7$. Thus, the number N of jobs that arrive in a given three-minute interval is a Poisson random variable, with $\mathbf{E}[N] = 3\lambda = 21$, $\text{var}(N) = 21$, and PMF

$$p_N(n) = \frac{(21)^n e^{-21}}{n!}, \quad n = 0, 1, \dots$$

Solution:

(b) Each of these 10 jobs has probability $\lambda_A/(\lambda_A + \lambda_B) = 3/7$ of being of type A, independently of the others. Thus, the binomial PMF applies and the desired probability is equal to

$$\binom{10}{3} \left(\frac{3}{7}\right)^3 \left(\frac{4}{7}\right)^7.$$